

Basic Biochemistry

FOUNDATIONAL CONCEPTS

Biochemistry is the scientific discipline that explores the chemical processes and molecular interactions occurring within living organisms. It bridges biology and chemistry, providing the molecular framework for understanding life at its most fundamental level. From the structure of biomolecules to the enzymatic reactions that power cellular metabolism, biochemistry forms the bedrock of modern biological science and medicine.

This study guide covers the essential topics in basic biochemistry, enzymology, molecular biology, genetics, developmental biology, evolution, biotechnology, computational biology, and reproductive biology. Each section provides detailed notes, visual aids, and key concepts to support undergraduate and early graduate students in mastering these foundational areas of the life sciences.

Biomolecules: The Building Blocks of Life

BASIC BIOCHEMISTRY

Biomolecules are the organic compounds produced by living organisms that are essential for biological processes. They are broadly classified into four major categories: carbohydrates, proteins, lipids, and nucleic acids. Each class plays distinct and vital roles in cellular structure, energy storage, signaling, and information transfer. Understanding the structure and function of these molecules is the first step toward comprehending the chemistry of life.

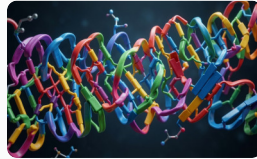


Carbohydrates

Carbohydrates are polyhydroxy aldehydes or ketones with the general formula $(CH_2O)_n$. They serve as the primary energy source for cells and also play structural roles.

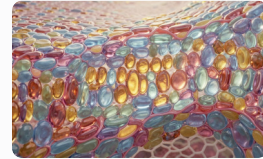
Monosaccharides (e.g., glucose, fructose) are the simplest units; disaccharides (e.g., sucrose, lactose) form from two monosaccharides; and polysaccharides (e.g., starch, glycogen, cellulose) are long chains serving as energy stores or structural components.

Glycosidic bonds link monosaccharide units together through condensation reactions.



Proteins

Proteins are polymers of amino acids linked by peptide bonds. They exhibit four levels of structure: primary (amino acid sequence), secondary (α -helices and β -sheets), tertiary (3D folding), and quaternary (multiple subunits). Proteins function as enzymes, structural components, transporters, hormones, and antibodies. The 20 standard amino acids differ in their side chains (R-groups), which determine protein properties and folding patterns.



Lipids

Lipids are a diverse group of hydrophobic or amphipathic molecules including fats, phospholipids, steroids, and waxes. Triglycerides (three fatty acids + glycerol) store energy efficiently. Phospholipids form the lipid bilayer of cell membranes. Steroids (e.g., cholesterol, hormones) serve signaling and structural roles. Saturated fatty acids lack double bonds; unsaturated fatty acids contain one or more double bonds, affecting membrane fluidity and physical properties.

Key Functions Summary

- Carbohydrates → Energy & structure
- Proteins → Catalysis, transport, signaling
- Lipids → Membranes, energy storage, hormones
- Nucleic Acids → Genetic information storage

Important Bonds to Know

- **Glycosidic bond** — carbohydrates
- **Peptide bond** — proteins
- **Ester bond** — lipids
- **Phosphodiester bond** — nucleic acids
- **Hydrogen bonds** — stabilize all structures

Enzymology & Metabolism

ENZYMES · KINETICS · CENTRAL METABOLISM

Enzymes are biological catalysts — almost always proteins — that accelerate chemical reactions in living systems without being consumed in the process. They lower the activation energy of reactions, enabling biochemical processes to occur at physiologically relevant rates. Enzyme activity is influenced by temperature, pH, substrate concentration, and the presence of inhibitors or activators. Understanding enzyme kinetics is essential for drug design, metabolic engineering, and clinical diagnostics.

Enzyme Kinetics: Michaelis-Menten Model

The Michaelis-Menten equation describes the relationship between reaction velocity (V) and substrate concentration $[S]$:

$$V = \frac{V_{max}[S]}{K_m + [S]}$$

Where **V_{max}** is the maximum velocity and **K_m** (Michaelis constant) is the substrate concentration at half V_{max} — a measure of enzyme-substrate affinity. A low K_m indicates high affinity. Competitive inhibitors increase K_m ; non-competitive inhibitors decrease V_{max} . Lineweaver-Burk plots (double-reciprocal plots) are used to determine these parameters graphically.

📌 Competitive inhibition can be overcome by increasing substrate concentration; non-competitive inhibition cannot.

Types of Enzyme Inhibition

Competitive

Inhibitor binds active site; competes with substrate. V_{max} unchanged, K_m increases.

Non-Competitive

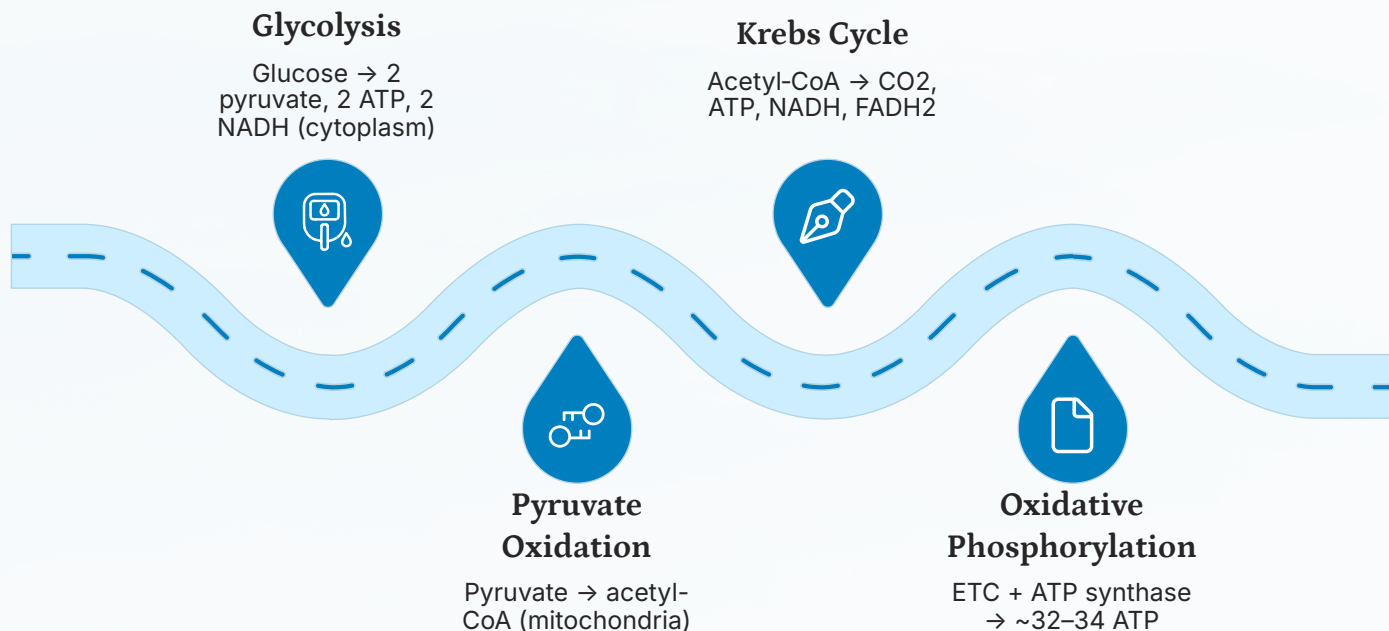
Binds allosteric site; reduces V_{max} . K_m unchanged.

Uncompetitive

Binds only enzyme-substrate complex; both V_{max} and K_m decrease.

Central Metabolic Pathways

Cellular metabolism encompasses all chemical reactions that sustain life. The three central catabolic pathways — glycolysis, the Krebs cycle, and oxidative phosphorylation — work in concert to extract energy from glucose and convert it into ATP, the universal energy currency of the cell.



Glycolysis occurs in the cytoplasm and converts one glucose molecule into two pyruvate molecules, producing a net gain of 2 ATP and 2 NADH. It consists of an energy-investment phase (steps 1–5) and an energy-harvesting phase (steps 6–10). Key regulatory enzymes include hexokinase, phosphofructokinase-1 (the major control point), and pyruvate kinase.

The **Krebs Cycle** (Citric Acid Cycle) takes place in the mitochondrial matrix. Acetyl-CoA enters the cycle by combining with oxaloacetate to form citrate. Through eight enzymatic steps, two carbons are released as CO₂, and energy is captured as 3 NADH, 1 FADH₂, and 1 GTP/ATP per turn. The cycle also provides intermediates for biosynthesis (anaplerotic reactions).

Oxidative Phosphorylation occurs at the inner mitochondrial membrane. NADH and FADH₂ donate electrons to the electron transport chain (Complexes I–IV), creating a proton gradient across the membrane. ATP synthase (Complex V) uses this gradient to synthesize ATP through chemiosmosis. Oxygen serves as the final electron acceptor, forming water. This stage produces approximately 32–34 ATP per glucose molecule.

Molecular Biology: DNA, RNA & the Central Dogma

MOLECULAR BIOLOGY

Molecular biology is the study of the molecular basis of biological activity, focusing on the interactions between DNA, RNA, and proteins. The Central Dogma of Molecular Biology, proposed by Francis Crick, describes the flow of genetic information: **DNA → RNA → Protein**. This framework underlies all genetic processes in living organisms and is fundamental to understanding gene expression, heredity, and biotechnology applications.



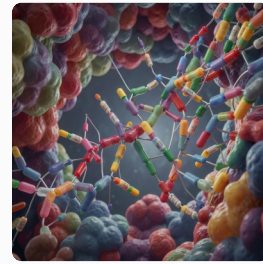
DNA Replication

DNA replication is semi-conservative — each daughter molecule contains one parental strand and one newly synthesized strand. Key enzymes include **helicase** (unwinds DNA), **primase** (synthesizes RNA primers), **DNA polymerase III** (main synthesis enzyme), **DNA polymerase I** (removes primers), and **ligase** (joins Okazaki fragments). Replication proceeds 5'→3' on both strands; the leading strand is continuous while the lagging strand is discontinuous.



Transcription

Transcription is the synthesis of RNA from a DNA template by **RNA polymerase**. In prokaryotes, a single RNA polymerase handles all transcription. In eukaryotes, RNA Pol I, II, and III transcribe rRNA, mRNA, and tRNA respectively. The process involves initiation (promoter binding), elongation (RNA synthesis 5'→3'), and termination. Eukaryotic pre-mRNA undergoes processing: 5' capping, 3' polyadenylation, and splicing to remove introns.



Translation

Translation is the synthesis of proteins from mRNA at ribosomes. Transfer RNA (tRNA) molecules carry specific amino acids and recognize mRNA codons through anticodon pairing. Translation occurs in three stages: **initiation** (ribosome assembly at start codon AUG), **elongation** (amino acid chain growth), and **termination** (release at stop codon). The genetic code is universal, degenerate (multiple codons per amino acid), and non-overlapping.

The Genetic Code

64 codons encode 20 amino acids + 3 stop signals. Key features:

- Universal across almost all organisms
- Degenerate — multiple codons per amino acid
- Non-overlapping and comma-free
- AUG = start codon (Methionine)
- UAA, UAG, UGA = stop codons

Key Enzymes & Factors

- **DNA Polymerase** — replication
- **RNA Polymerase** — transcription
- **Helicase** — unwinds DNA
- **Ligase** — joins DNA fragments
- **Topoisomerase** — relieves supercoiling
- **Reverse Transcriptase** — RNA→DNA

Gene Regulation: Operons & Control Mechanisms

GENE REGULATION

Gene regulation refers to the mechanisms that control when and how much a gene is expressed. In prokaryotes, genes are often organized into **operons** — clusters of functionally related genes under the control of a single promoter. The operon model, proposed by Jacob and Monod (1961), was the first detailed explanation of gene regulation and remains a cornerstone of molecular biology. Eukaryotic gene regulation is more complex, involving chromatin remodeling, transcription factors, and epigenetic modifications.

The Lac Operon Model

The **lac operon** in *E. coli* controls the metabolism of lactose. It consists of three structural genes (**lacZ**, **lacY**, **lacA**), an operator (O), a promoter (P), and a regulatory gene (**lacI**) that produces the repressor protein. When lactose is absent, the repressor binds the operator, blocking transcription. When lactose is present, allolactose (an isomer) binds the repressor, causing it to release the operator and allowing transcription. Additionally, **CAP-cAMP** complex enhances transcription when glucose is low — a classic example of catabolite repression.

❑ **Inducible system:** Normally OFF; turned ON by an inducer (lactose/allolactose).

Trp Operon — Repressible System

The **trp operon** controls tryptophan biosynthesis and operates oppositely to the lac operon. When tryptophan is abundant, it acts as a **corepressor**, binding to the inactive repressor and activating it to block transcription. When tryptophan is scarce, the repressor remains inactive and transcription proceeds. This is an example of **feedback repression** — the end product of a pathway regulates its own synthesis.

❑ **Repressible system:** Normally ON; turned OFF by a corepressor (tryptophan).

Levels of Gene Regulation

01

Transcriptional Control

Regulation of when and how often a gene is transcribed — the most common level. Involves promoters, enhancers, silencers, and transcription factors.

03

Translational Control

Regulation of ribosome binding and translation initiation rates controls protein synthesis from existing mRNA.

02

Post-Transcriptional Control

Alternative splicing, mRNA stability, and RNA interference (miRNA, siRNA) regulate mRNA processing and lifespan.

04

Post-Translational Control

Protein modification (phosphorylation, glycosylation, ubiquitination) and degradation regulate protein activity and abundance.

Classical & Population Genetics

GENETICS

Genetics is the study of heredity and the variation of inherited characteristics. Classical genetics, founded by Gregor Mendel's experiments with pea plants (1865), established the fundamental laws of inheritance. Population genetics extends these principles to understand how allele frequencies change in populations over time, providing the mathematical foundation for evolutionary theory.

Mendel's Laws

- **Law of Segregation** — Alleles separate during gamete formation; each gamete receives one allele per gene.
- **Law of Independent Assortment** — Genes on different chromosomes assort independently during meiosis.
- **Law of Dominance** — In heterozygotes, the dominant allele masks the recessive allele's expression.

Monohybrid cross ratio: **3:1** (phenotypic)

Dihybrid cross ratio: **9:3:3:1**

Linkage & Crossing Over

Genes located on the same chromosome tend to be inherited together — this is **genetic linkage**. The frequency of **crossing over** (recombination) between linked genes is proportional to the physical distance between them. One map unit (centimorgan, cM) = 1% recombination frequency. Thomas Hunt Morgan's work with *Drosophila* established the chromosomal theory of inheritance and linkage mapping.

Mutations & Chromosomal Aberrations

- **Point mutations:** Silent, missense, nonsense, frameshift
- **Chromosomal:** Deletion, duplication, inversion, translocation
- **Aneuploidy:** Trisomy (2n+1), Monosomy (2n-1)
- **Polyploidy:** 3n, 4n — common in plants

Hardy–Weinberg Equilibrium

The Hardy-Weinberg principle states that allele and genotype frequencies in a population will remain constant from generation to generation in the absence of evolutionary influences. It provides a null model for detecting evolutionary change.

$$p^2 + 2pq + q^2 = 1 \quad \text{and} \quad p + q = 1$$

Where **p** = frequency of dominant allele, **q** = frequency of recessive allele, **p²** = homozygous dominant, **2pq** = heterozygous, **q²** = homozygous recessive. Equilibrium requires: large population, random mating, no mutation, no migration, and no natural selection.

Population Genetics Forces

- Natural Selection
- Genetic Drift
- Gene Flow
- Mutation
- Non-random Mating

Types of Selection

- Directional — favors one extreme
- Stabilizing — favors intermediate
- Disruptive — favors both extremes
- Sexual — mate choice driven

Key Concepts

- Gene pool & allele frequency
- Founder effect & bottleneck
- Inbreeding coefficient
- Heterozygote advantage

Developmental Biology

EMBRYOLOGY & DEVELOPMENT

Developmental biology studies the processes by which a single fertilized egg (zygote) develops into a complex multicellular organism. It encompasses cell differentiation, morphogenesis, growth, and pattern formation. Understanding developmental biology is crucial for regenerative medicine, stem cell research, and understanding congenital disorders. The field integrates molecular biology, genetics, and cell biology to explain how genetic information is translated into form and function.



Cleavage & Blastulation

Cleavage is the rapid series of mitotic divisions following fertilization, converting the zygote into a multicellular structure without increasing overall size. Cleavage patterns vary: *holoblastic* (complete, in isolecithal eggs like mammals) vs. *meroblastic* (partial, in telolecithal eggs like birds). The result is a solid ball of cells called the **morula**, which then forms a hollow sphere — the **blastula** — with a fluid-filled cavity called the blastocoel.



Gastrulation & Germ Layers

Gastrulation is the dramatic reorganization of the blastula into a three-layered structure called the **gastrula**. Three primary germ layers are established: **ectoderm** (outer — skin, nervous system), **mesoderm** (middle — muscles, bones, circulatory system), and **endoderm** (inner — gut lining, respiratory tract). Key morphogenetic movements include invagination, involution, epiboly, and delamination.



Organizer Concept & Embryonic Induction

The **Spemann-Mangold organizer** (dorsal lip of the blastopore in amphibians) is a group of cells that can induce the formation of a complete embryonic axis when transplanted. This discovery demonstrated **embryonic induction** — the process by which one group of cells (the inducer) influences the developmental fate of neighboring cells (the responder) through signaling molecules like BMP, Wnt, and Nodal. This concept is fundamental to understanding pattern formation.



Metamorphosis

Metamorphosis is a profound post-embryonic transformation involving dramatic changes in body structure and function. In amphibians, thyroid hormone (thyroxine) triggers the transition from aquatic larva (tadpole) to terrestrial adult. This process involves programmed cell death (apoptosis), tissue remodeling, and organ regeneration. In insects, metamorphosis is controlled by ecdysone and juvenile hormone, with holometabolous insects undergoing complete transformation through pupal stages.

Evolutionary Biology

EVOLUTION

Evolutionary biology is the study of the processes that have produced the diversity of life on Earth over billions of years. Evolution is defined as a change in the heritable characteristics of biological populations over successive generations. The modern synthesis integrates Darwinian natural selection with Mendelian genetics, population genetics, and molecular biology to provide a comprehensive framework for understanding the history and mechanisms of evolution.

Darwin's Theory of Natural Selection

Charles Darwin proposed natural selection as the primary mechanism of evolution in *On the Origin of Species* (1859). The theory rests on four observations:

1. Organisms produce more offspring than can survive
2. Individuals vary in heritable traits
3. Some traits confer survival/reproductive advantage
4. Advantageous traits become more common over generations

Fitness = relative reproductive success. Natural selection acts on phenotypes, but evolution occurs at the population level through changes in allele frequencies.

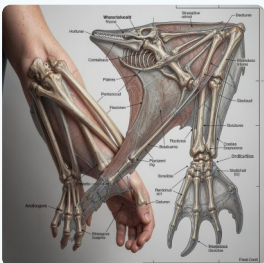
Speciation & Adaptive Radiation

Speciation is the formation of new and distinct species. Key modes include:

- **Allopatric** — geographic isolation (most common)
- **Sympatric** — without geographic separation
- **Parapatric** — adjacent populations with limited gene flow
- **Peripatric** — small isolated population at range edge

Adaptive radiation is the rapid diversification of a lineage into multiple ecological niches (e.g., Darwin's finches, Hawaiian honeycreepers, cichlid fishes). It often follows mass extinctions or colonization of new environments.

Evidence & Concepts in Evolution



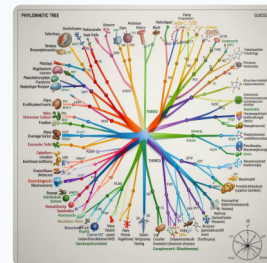
Homologous vs. Analogous Organs

Homologous structures share a common evolutionary origin but may differ in function (e.g., human arm, bat wing, whale flipper — same bone pattern). They indicate **divergent evolution**. **Analogous structures** have similar functions but different origins (e.g., bird wing vs. insect wing). They result from **convergent evolution** — independent adaptation to similar environmental pressures.



Evolutionary Theories

Beyond Darwinism, several theories have shaped evolutionary thought: **Lamarckism** (inheritance of acquired characteristics — largely discredited), **Neo-Darwinism** (modern synthesis with genetics), **Punctuated Equilibrium** (Eldredge & Gould — long stasis punctuated by rapid change), and **Neutral Theory** (Kimura — most molecular changes are selectively neutral). Endosymbiotic theory explains the origin of mitochondria and chloroplasts from ancient prokaryotic engulfment events.



Phylogenetics & Systematics

Phylogenetic trees represent evolutionary relationships among species based on shared derived characteristics (synapomorphies). **Cladistics** groups organisms by common ancestry. Molecular phylogenetics uses DNA/protein sequence data to reconstruct evolutionary history. Key terms: **monophyletic** (clade), **paraphyletic**, and **polyphyletic** groups. Molecular clocks estimate divergence times based on mutation rates.

Biotechnology & Computational Biology

BIOTECHNOLOGY

COMPUTATIONAL BIOLOGY

Biotechnology applies biological systems, living organisms, or their derivatives to develop products and technologies that improve human life. From recombinant DNA technology to CRISPR-Cas9 gene editing, modern biotechnology has revolutionized medicine, agriculture, and industry. Computational biology and bioinformatics provide the analytical tools to manage and interpret the vast amounts of biological data generated by modern research.

Recombinant DNA Technology

Recombinant DNA (rDNA) technology involves combining DNA from different sources to create novel genetic combinations. Key tools include:

- **Restriction enzymes** — molecular scissors that cut DNA at specific sequences
- **DNA ligase** — joins DNA fragments
- **Cloning vectors** — plasmids, bacteriophages, cosmids, BACs, YACs
- **Host cells** — *E. coli*, yeast, mammalian cells

PCR (Polymerase Chain Reaction)

PCR amplifies specific DNA sequences exponentially through repeated cycles of: **denaturation** (94–95°C), **annealing** (50–65°C), and **extension** (72°C). Taq polymerase (thermostable) is used. Applications include diagnostics, forensics, and research. Variants: RT-PCR, qPCR, nested PCR.

Key Biotech Techniques

- **Gel Electrophoresis** — separates DNA/RNA/proteins by size using electric field through agarose or polyacrylamide gel
- **Southern Blot** — detects specific DNA sequences
- **Northern Blot** — detects RNA
- **Western Blot** — detects proteins using antibodies
- **Animal Cell Culture** — in vitro growth of animal cells for research, vaccine production, and biopharmaceuticals

CRISPR-Cas9 & Computational Biology



CRISPR-Cas9

A revolutionary gene-editing tool derived from bacterial immune systems. Cas9 nuclease, guided by sgRNA, creates double-strand breaks at specific genomic locations. Applications include gene knockout, knock-in, base editing, and epigenetic modification. Ethical considerations surround germline editing.



Computational Biology

Uses computational methods to analyze biological data. Includes genome assembly, sequence alignment (BLAST), phylogenetic analysis, protein structure prediction (AlphaFold), and systems biology modeling. Bioinformatics databases: GenBank, PDB, UniProt.



Biostatistics

Statistical methods applied to biological data. Key concepts: descriptive statistics, hypothesis testing (t-test, chi-square, ANOVA), regression analysis, probability distributions, and experimental design. Essential for interpreting clinical trials and population studies.



Research Methodology

Systematic approach to scientific investigation: hypothesis formulation, experimental design, controls (positive/negative), replication, data collection, statistical analysis, and peer-reviewed publication. Good Laboratory Practice (GLP) ensures reproducibility and ethical standards.

Endocrinology & Reproductive Biology

ENDOCRINOLOGY

REPRODUCTIVE BIOLOGY

Endocrinology is the study of hormones and the endocrine glands that secrete them. Hormones are chemical messengers that regulate physiology, growth, metabolism, and reproduction by acting on target cells through specific receptors. Reproductive biology examines the mechanisms of gamete formation, fertilization, and the hormonal control of reproductive cycles. Together, these fields are essential for understanding human health, fertility, and the development of therapeutic interventions.

Hormones & Endocrine Glands

The endocrine system includes glands that secrete hormones directly into the bloodstream. Major glands and their hormones:

- **Hypothalamus** — releasing/inhibiting hormones; controls pituitary
- **Pituitary** — GH, TSH, ACTH, FSH, LH, prolactin, ADH, oxytocin
- **Thyroid** — T3, T4 (metabolism); Calcitonin (Ca^{2+} regulation)
- **Parathyroid** — PTH (raises blood Ca^{2+})
- **Adrenal** — cortisol (stress), aldosterone (Na^+/K^+), adrenaline
- **Pancreas** — insulin (lowers blood glucose), glucagon (raises glucose)
- **Gonads** — testosterone, estrogen, progesterone

Hormones act via two main receptor types: **steroid hormones** (lipid-soluble, intracellular receptors, alter gene expression) and **peptide hormones** (water-soluble, membrane receptors, second messenger systems like cAMP).

Contraceptive Devices

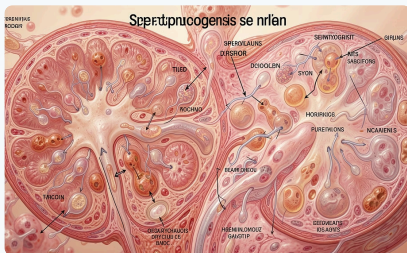
Contraception prevents pregnancy through various mechanisms:

- **Barrier methods** — condoms, diaphragms (block sperm)
- **Hormonal** — oral pills, patches, implants (prevent ovulation)
- **IUD** — intrauterine devices (copper or hormonal)
- **Emergency contraception** — prevents implantation
- **Permanent** — vasectomy, tubal ligation

Key Regulatory Axes

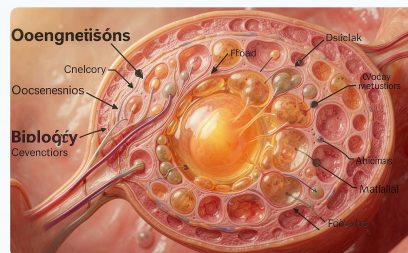
- **HPG Axis** — Hypothalamus → Pituitary → Gonads
- **HPA Axis** — Hypothalamus → Pituitary → Adrenals
- **HPT Axis** — Hypothalamus → Pituitary → Thyroid

Gametogenesis: Spermatogenesis & Oogenesis



Spermatogenesis

Spermatogenesis is the formation of sperm in the seminiferous tubules of testes. It begins at puberty and continues throughout life. Process: **spermatogonia** ($2n$) → mitosis → **primary spermatocytes** ($2n$) → Meiosis I → **secondary spermatocytes** (n) → Meiosis II → **spermatids** (n) → **spermatozoa** (spermiogenesis). FSH stimulates Sertoli cells; LH stimulates Leydig cells to produce testosterone. One spermatogonium yields **4 functional sperm**.



Oogenesis

Oogenesis is the formation of ova (eggs) in the ovaries. It begins before birth (primary oocytes arrested in prophase I) and completes only upon fertilization. Process: **oogonia** → **primary oocyte** (arrested until puberty) → Meiosis I → **secondary oocyte + 1st polar body** → Meiosis II (completed at fertilization) → **ovum + 2nd polar body**. One oogonium yields **1 functional ovum** and 3 polar bodies. The menstrual cycle (~28 days) is regulated by FSH, LH, estrogen, and progesterone through the HPG axis.

- ✔ **Key Comparison:** Spermatogenesis produces 4 sperm continuously from puberty; Oogenesis produces 1 egg per cycle with long meiotic arrests. Both are regulated by the hypothalamic-pituitary-gonadal (HPG) axis through feedback loops.